

2.8 Kinetics

Students will be assessed on their ability to:

- a recall the factors that influence the rate of chemical reaction, including concentration, temperature, pressure, surface area and catalysts
- b explain the changes in rate based on a qualitative understanding of collision theory
- c use, in a qualitative way, the Maxwell-Boltzmann model of the distribution of molecular energies to relate changes of concentration and temperature to the alteration in the rate of a reaction
- d demonstrate an understanding of the concept of activation energy and its qualitative relationship to the effect of temperature changes on the rate of reaction
- e demonstrate an understanding of the role of catalysts in providing alternative reaction routes of lower activation energy and draw the reaction profile of a catalysed reaction including the energy level of the intermediate formed with the catalyst
- f carry out simple experiments to demonstrate the factors that influence the rate of chemical reactions, e.g. the decomposition of hydrogen peroxide.